

ShopEZ Test Validation Report

This report summarizes the verification results and test execution suites performed for the **ShopEZ** application, satisfying all QA standards.

1. Test Summary Overview

- **Total Tests**: 6
- **Passed**: 6
- **Failed**: 0
- **Skipped**: 0
- **Execution Engine**: Node-native assertion framework (`node tests/run_tests.js`)

2. Test Execution Details

Test 1: JWT Signature & Verification

- **Objectives**: Validate payload serialization, secret hashing, and token decryption.
- **Outcome**: Tokens are generated and validated correctly. Decoded user parameters match signed inputs.
- **Status**: **PASS**

Test 2: Auth Middleware Gate Check

- **Objectives**: Verify that valid Bearer tokens pass to route handlers and invalid or missing headers are blocked with a `401 Unauthorized` status.
- **Outcome**: Confirmed. Requests without a valid token are rejected.
- **Status**: **PASS**

Test 3: Javascript Fallback Recommendation Engine

- **Objectives**: Verify that in-memory TF-IDF and cosine similarity return meaningful related products if the Flask server is down.
- **Outcome**: A test catalog was processed; related products matching term weight similarity were correctly returned.
- **Status**: **PASS**

Test 4: Administrator Middleware Check (RBAC Gate)

- **Objectives**: Verify that requests from non-administrator users to admin routes are blocked with a `403 Forbidden` response.
- **Outcome**: Stubb-tested with mock admin and customer entities. Checked MongoDB query resolution.
- **Status**: **PASS**

Test 5: Checkout Card Input Verification

- **Objectives**: Verify checking functions on card digits, CVV lengths, expiry formats (MM/YY), and cardholder name validations.
- **Outcome**: Standard invalid inputs are correctly caught, and valid card numbers pass validation.
- **Status**: **PASS**

Test 6: Checkout UPI ID Format Verification

- **Objectives**: Verify that user-entered UPI handles contain the `@` separator (e.g. `user@okbank`).
- **Outcome**: Confirmed. Non-conforming string formats are blocked.
- **Status**: **PASS**

3. Execution Log Output

```
=====
SHOPEZ UNIT & INTEGRATION TESTS
=====
■ [Test 1] Testing JWT Signing and Verification...
✓ [Test 1] Passed successfully.
■ [Test 2] Testing authMiddleware gate...
✓ [Test 2] Passed successfully.
■ [Test 3] Testing JS native TF-IDF Recommendation algorithm...
✓ [Test 3] Passed successfully.
■ [Test 4] Testing adminMiddleware gate...
✓ [Test 4] Passed successfully.
■ [Test 5] Testing card input validator logic...
✓ [Test 5] Passed successfully.
■ [Test 6] Testing UPI input validator logic...
✓ [Test 6] Passed successfully.

■ ALL 6 TEST SUITES PASSED SUCCESSFULLY.
=====
```

4. Defect Analysis & Bug Log

For details on bugs discovered and resolved during our development sprints, refer to the Bug Reports Matrix.